

Class Concepts

Definition and Core Concepts of a Class

Class Definition

A class is a blueprint for creating objects. It groups data (attributes) and functions (methods) together. A class does not use memory on its own; only an object made from it does.

Example:

A blank admission form is like a class. A filled form for an actual student is the object.

Code Example:

```
class Student {
private:
    string name;  int age;

public:

    Student(string n, int a) {

        name = n;

        age = a; }

    void setAge(int a) {

        if (a > 0) age = a; }

    void showDetails() {

        cout << name << ", " << age << endl; }

    ~Student() {

        cout << "Record deleted." << endl; }

int main() {
    string userName;
    int userAge;
    cin >> userName >> userAge;
```

```
        Student s1(userName, userAge);  
s1.showDetails();  
        s1.setAge(21);  
        s1.showDetails();  
return 0;  
}
```

Core Concepts of a Class

Attributes

Variables that store an object's properties.

Example:

Color and speed are attributes of a car.

Methods

Functions that define what an object can do.

Example:

Start and brake are methods of a car.

Access Specifiers

Control visibility: private (class only), public (anywhere), protected (class and child classes).

Example:

A diary is private, a notice board is public, a family matter is protected.

Constructor

Runs automatically when an object is created, used to set initial values.

Example:

Opening an account auto-sets the account number and starting balance.

Destructor

Runs automatically when an object is destroyed, used for cleanup.

Example:

Closing an account automatically settles and shuts it down.

Encapsulation

Hiding data as private, accessible only through public methods.

Example:

*A remote hides its wiring; you only use
the outer buttons.*

Code of Calculator using class

```
Welcome Calculator.cpp
Calculator.cpp > Calculator
1  #include <iostream>
2  using namespace std;
3
4  class Calculator {
5      private:
6          int a, b;
7
8      public:
9          // Constructor
10         Calculator(int a, int b) {
11             this->a = a;
12             this->b = b;
13         }
14
15         // Setters
16         void setA(int a) {
17             this->a = a;
18         }
19
20         void setB(int b) {
21             this->b = b;
22         }
```



```
welcome Calculator.cpp
Calculator.cpp > Calculator
4  class Calculator {
23
24      // Show current values
25      void showDetails() {
26          cout << "a = " << a << ", b = " << b << endl;
27      }
28
29      // Operations
30      void add() {
31          cout << "Addition: " << a + b << endl;
32      }
33
34      void subtract() {
35          cout << "Subtraction: " << a - b << endl;
36      }
37
38      void multiply() {
39          cout << "Multiplication: " << a * b << endl;
40      }
41
42      void divide() {
43          if (b == 0) {
44              cout << "Error: Division by zero" << endl;
45          } else {
46              cout << "Division: " << a / b << endl;
47          }
48      }
49
50      // Destructor
51      ~Calculator() {
52          cout << "Calculator object deleted." << endl;
53      }
54  };
55
Ln 9, Col 17 (86 selected) Spaces: 4 UTF-8 CRLF {
```

```

5
6  int main() {
7      Calculator c1(20, 40);
8      c1.showDetails();
9
10     c1.add();
11     c1.subtract();
12     c1.multiply();
13     c1.divide();
14
15     c1.setA(45);
16     c1.showDetails();
17
18     return 0;
19 }

```

```

PS C:\Users\AriEs AngEL\Downloads\Calculator.cpp -o Calculator.exe
PS C:\Users\AriEs AngEL\Downloads\Calculator.exe
a = 20, b = 40
Addition: 60
Subtraction: -20
Multiplication: 800
Division: 0
a = 45, b = 40
Calculator object deleted.
PS C:\Users\AriEs AngEL\Downloads\Calculator

```

